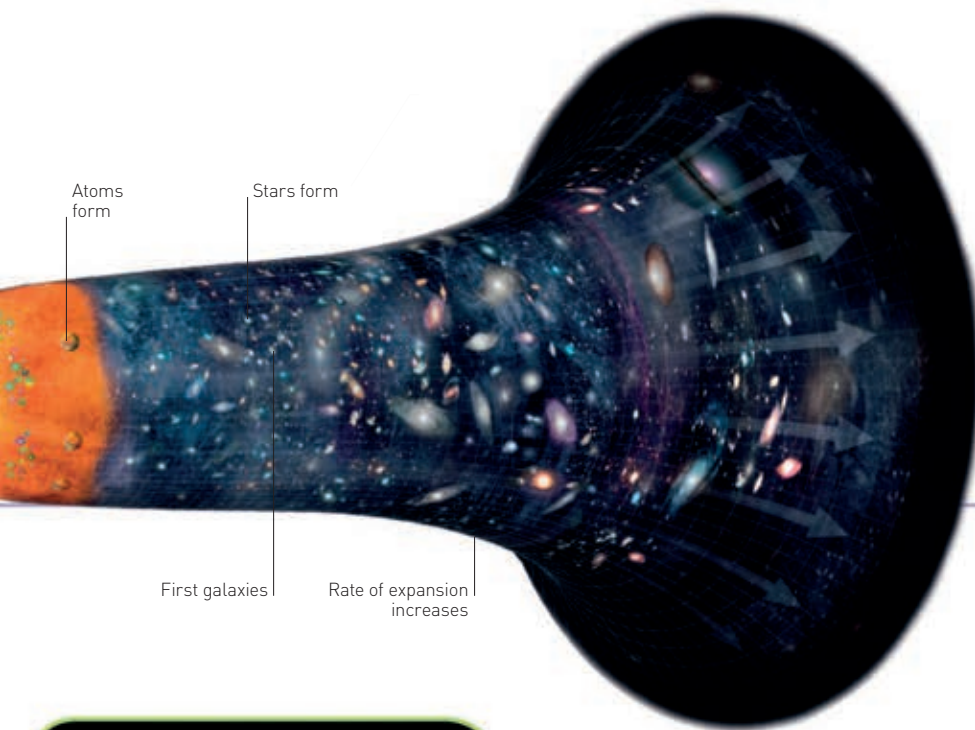
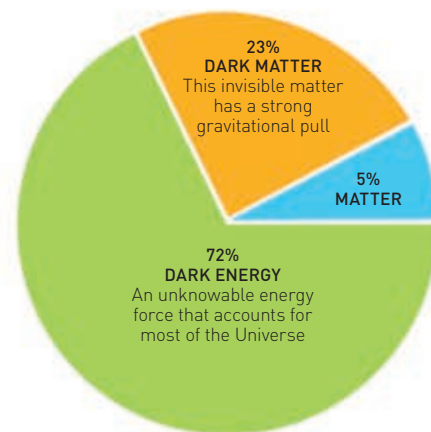




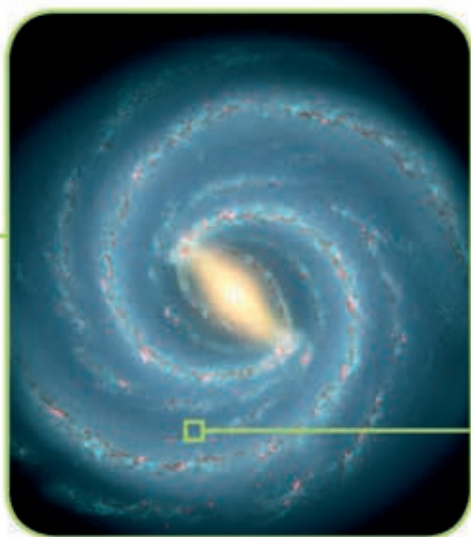
WHAT MAKES UP THE UNIVERSE?

The Universe contains matter and energy. Matter is generally physical "stuff" that can be seen, like the planets, but galaxies also contain invisible matter called "dark matter". This does not give off light or heat and so can be detected only by the effects of its gravity on visible objects. Between and beyond both types of matter is "dark energy", a mysterious thing that scientists know almost nothing about.



THE SCALE OF THE UNIVERSE

The Universe is so vast that it is hard to appreciate its size. This series of pictures "zooms in" on the Universe, to show how our Solar System and planet relate to the rest of the Universe. Space is so huge that astronomers use the speed of light to measure distances. One light year is the distance light travels in a year, which is nearly 10 trillion km (6 trillion miles).



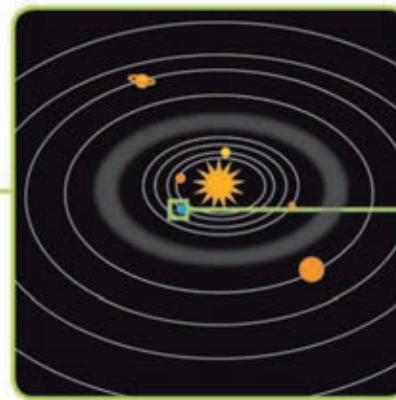
MILKY WAY GALAXY

The Milky Way has a spiral shape and holds around 200 billion stars within its gravitational pull.



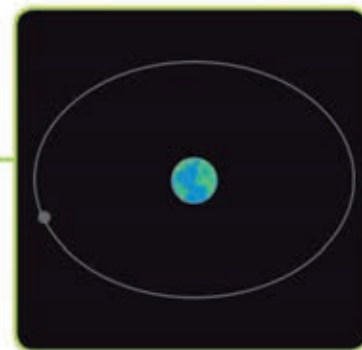
STELLAR NEIGHBOURHOOD

Our Solar System is on one of the Milky Way's spiral arms, 27,000 light years from the galaxy's centre.



SOLAR SYSTEM

The Sun sits at the centre of our Solar System, and eight major planets orbit it.



EARTH AND MOON

Earth is one of the planets orbiting the Sun, and the Moon orbits Earth.

BLACK HOLES

A black hole is a region of space where matter has collapsed in on itself. This means there is nothing to be seen, but astronomers know black holes exist because they have such a strong gravitational pull that nothing can escape them – not even light.



ARTIST'S IDEA OF A BLACK HOLE

DWARF PLANETS

Large planets have enough self-gravity to make them form into a round shape as they move through space. Smaller planets that cannot do this, but do orbit the Sun, are called "dwarf planets". Pluto is one of the largest dwarf planets in our solar system.



PLUTO

COMETS

Comets are small, icy worlds that orbit the Sun. They are made of frozen gases, rock, and dust. As they orbit the Sun, jets of gas and dust vaporize behind them to create long "tails" visible in space.



COMET

PLANETS

Planets are large, spherical objects that orbit a star. In our Solar System, there are eight planets: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. Planets that occur outside our Solar System are known as exoplanets.



EARTH

MOONS

A moon is a rocky body that orbits a planet. Some planets have many moons but Earth has only one. Moons are also known as natural satellites.



EARTH'S MOON

ASTEROIDS

Asteroids are small rocky bodies that orbit the Sun. There are millions of them in space, and they are mainly made of materials that were left over from the formation of planets.



EROS, A NAMED ASTEROID